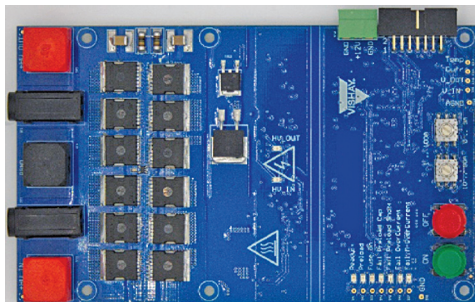


OF E-FUSE

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A Bidirectional E-Fuse



The reference design, equipped with SiC MOSFETs and a VOA300 optocoupler, handles up to 40kW continuous power for safe high-voltage power source connections in vehicles, such as battery packs.

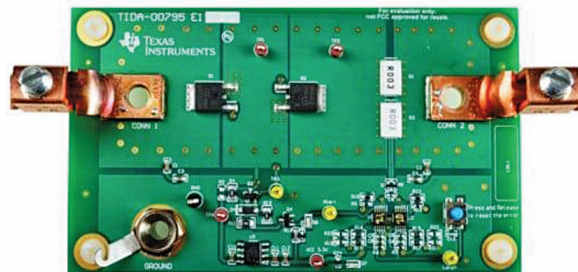
The bidirectional e-fuse reference design from Vishay is equipped with SiC MOSFETs and a VOA300 optocoupler designed to handle continuous power loads of up to 40kW. The design functions continuously at full power, experiencing minimal losses below 30W, and does not require active cooling. It is enriched with features like a preload function, continuous current monitoring, and swift overcurrent protection. Specifically crafted to control the connection and disconnection of high-voltage power sources, such as high-energy battery packs, to various vehicle loads, it uses SiC MOSFETs as its primary switches. These switches are capable of operating continuously at currents up to 100A. The e-fuse swiftly disconnects the load from the battery pack if the current goes beyond a set limit, ensuring the safety of both the battery pack and the user. It also detects excessive load capacitance or short circuits during power-up and quickly initiates shutdown. The high-voltage circuitry, connectors, control buttons, status LEDs, and multiple test points are on the top, while the low-voltage control is on the bottom. This e-fuse also offers remote control via a web browser.

The reference design is suitable for isolated bi-directional EV charging systems.

Vishay

<https://www.electronicsforu.com/electronics-projects/a-bidirectional-efuse-reference-design>

Automotive Precision E-Fuse



The reference design for an automotive precision e-fuse offers an advanced alternative to conventional fuses, boasting enhanced accuracy and additional features. The design is also utilised in the electric vehicle (EV) charging infrastructure.

The TIDA-00795 from Texas Instruments (TI) is a reference design for an automotive precision e-fuse offering an advanced alternative to conventional fuses, boasting enhanced accuracy and additional features. It is used in electric vehicle (EV) charging infrastructures where the human-machine interface (HMI) module augments electric vehicle charging station functions. This design's overcurrent limit reaches up to 30A, scalable beyond 100A, and boasts over 97% accuracy. It responds swiftly, peaking at 10μs but remains adjustable, and features a configurable delay time for inrush currents at intervals like 10, 50, or 100 μs. Enhanced accuracy is achieved via a configurable hysteresis feature, and error management offers power-off or push-button reset options. This automotive e-fuse design comprises specific subsections, including the current sense and activation component, which stands out in the e-fuse system with 90% consistency, making it adaptable for various current sense values. The design also encompasses the switch and control circuit and concludes with the power supply and protection section designed for transient suppression, accommodating automotive scenarios like jump starts, cold cranking, and reverse polarity.

The reference design lets users power their EVs using a household AC mains plug or private and public AC charging stations.

Texas Instruments (TI)

<https://www.electronicsforu.com/electronics-projects/reference-design-of-automotive-precision-efuse>